

STATE OF THE WORK · 29 SEPTEMBER 2026

F1R3Gaze

A browser whose only execution mechanism is f1r3lang on a native RSpace. Review of the proposal, the implementation specification and the code.



F1R3FLY.IO

WHERE THE WORK STANDS

THE SHORT VERSION

The specification is implemented through the signed installers and the reach tier. What remains is the graded resolver, the pieces that wait on node work, and the legacy and gateway tier.

13 crates

About 11,400 lines of Rust across the portable core, the native browser, the toolchain and the reach tier.

61 tests

Reported passing in the workspace, with 181 more in CampF1R3. Headless smoke test in CI on three OSes.

~35 of 62

Engineer-weeks of the plan's work packages delivered, by this review's count, before the plan's 5 October start.



THE DESIGN IN THREE IDENTIFICATIONS

FROM THE PROPOSAL, V0.2

A page is a process

Its behaviour is a f1r3lang program run by the CampF1R3 executive in the tab. No JavaScript is ever executed.

The document is a namespace

Elements are unforgeable names. A page can touch only the nodes it has been handed.

Events are data on channels

A click is a message. Handlers are receives, and the frame is the unit of commit.

what the web does	what f1r3gaze does instead
Same-origin policy	Capability discipline: a page imports named capabilities, granted and revocable by the broker
Unbounded scripts	Cost accounting as a per-page budget, enforced by the page meter
Event-loop ordering	Graded where-clauses as the programmable arbiter of interface races
Debugger breakpoints	Deterministic replay: identical commit hashes, frame by frame

THE STACK

layer	choice	status
Document engine	Blitz: Stylo, Taffy, Parley and Vello	Built
Execution	The CampF1R3 executive, one per tab	Built
Server	A F1R3FLY shard, through the shard bridge	Built to the quorum rung
Reach	The same executive as wasm in stock browsers	Built
Legacy web	Servo, quarantined; CEF as fallback	Not started

Electron and Tauri were rejected as the base: in both, JavaScript is the application runtime, so f1r3lang could only be a guest.



FROM PROPOSAL TO CODE

THREE DOCUMENTS IN TWO DAYS

28 SEP

Proposal v0.2

Reviews eight browser frameworks, picks Blitz, adds the shard as server and the rules for converting to and from the Web.



28 SEP

Specification v0.1

A line-by-line review of CampF1R3 finds four places the proposal assumed missing machinery.



29 SEP

F1R3Gaze v0.1.0

One commit, “full implementation”, with the upstream CampF1R3 changes carried as a patch.

The four gaps the specification found: no ground data in the kernel, guards parsed and discarded, no host channels or metering, and no oracle seam for choice. The first three are closed; the fourth is the graded resolver still to come.



WHAT IS BUILT

PORTABLE CORE AND REACH

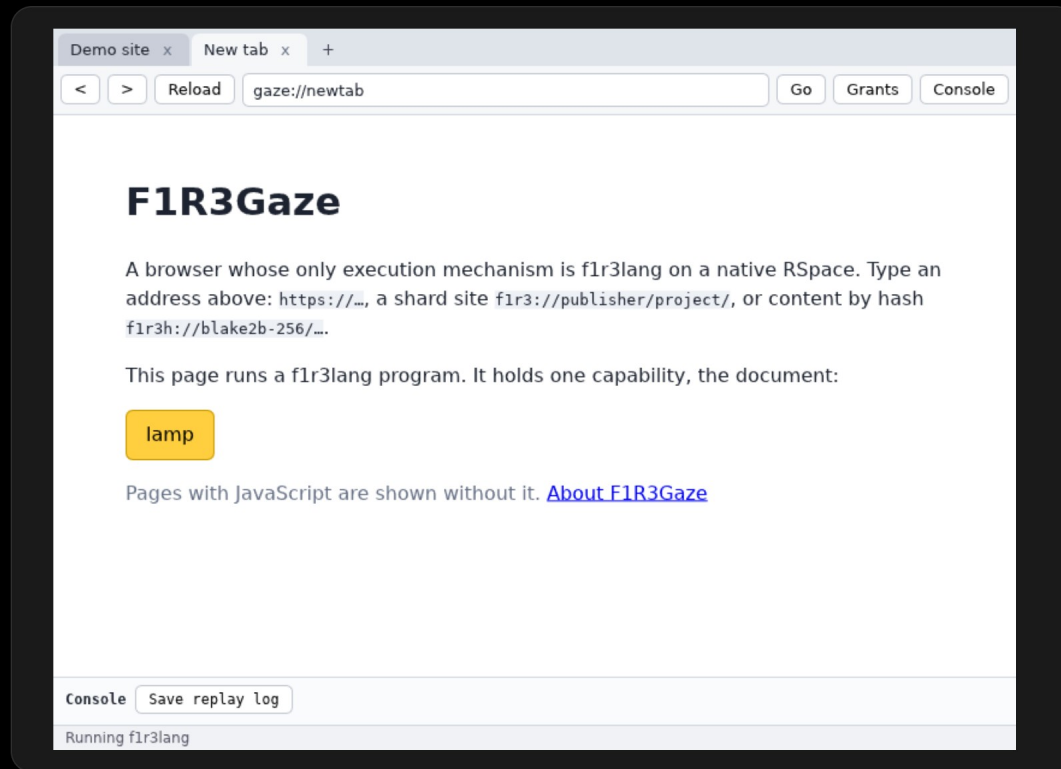
crate	spec	contents
gaze-graded	\$10	Four integer semirings, sampling, fairness, ceilings
gaze-knf	\$5	The .knf container, manifest, integrity strings
gaze-dom-core	\$7	DOM protocol engine, attenuation, events
gaze-exec	\$6, \$8	Tab executive, frame loop, meter, record and replay
gaze-broker	\$8.4	Grants, prompts, routes, revocation
gaze-reach	\$12	The reach tier as one wasm module

NATIVE BROWSER AND TOOLCHAIN

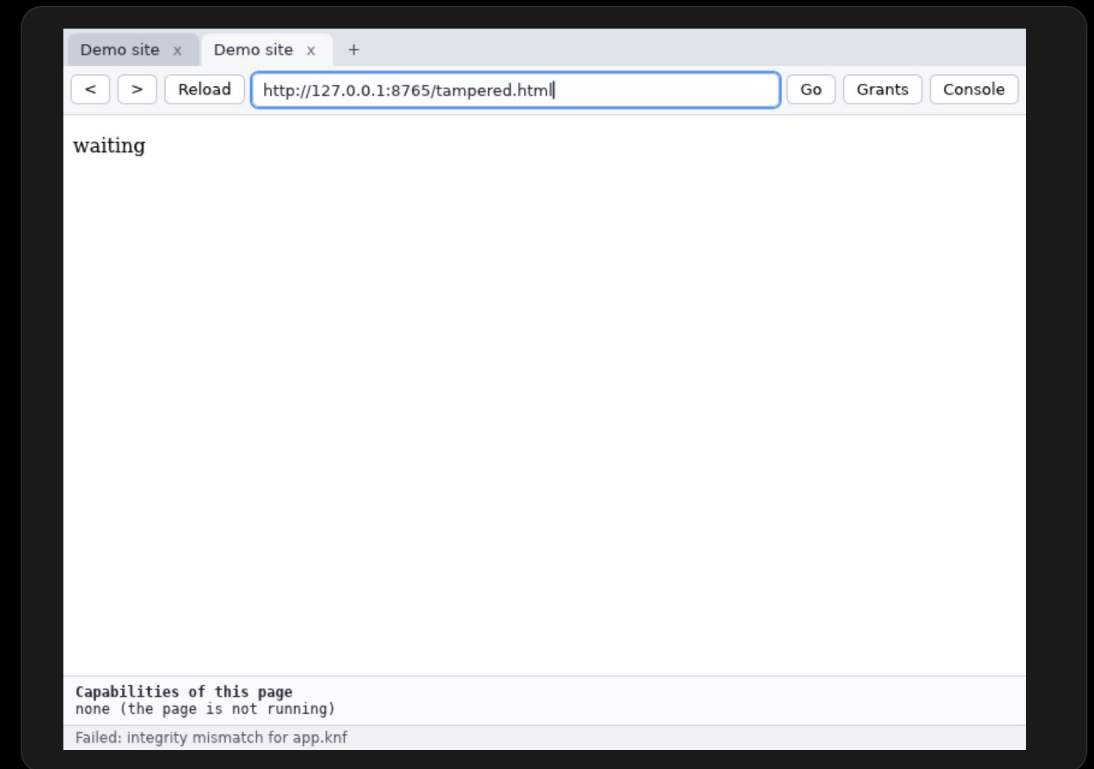
crate	spec	contents
gaze-dom-blitz	\$7.9	Blitz backend, RhoDocument, event handler
gaze-net	\$8.5	HTTP with no ambient credentials
gaze-store	\$8.6	Per-origin append-only store
gaze-blob	\$9.7	Content-addressed, verified blob cache
gaze-shard	\$9	Shard bridge: deploys, quorum, sessions, sites
gaze-shell	\$3	The application: tabs, chrome, headless runner
f1r3c	\$3	Toolchain: compile, inspect, integrity, site



RUNNING ON THE REAL BINARY



The new-tab page holds one capability, the document. A click lights the lamp.



A tampered .knf is refused: “integrity mismatch”, and the page gets no capabilities.



TRUST FROM CONTENT AND CONSENSUS

THE ASSURANCE LADDER: EVERY SHARD REPLY NAMES ITS RUNG

node	One observer's answer.	BUILT
quorum	Identical encodings from k of n observers run by distinct operators; 2 of 3 by default.	BUILT
proof	Signatures from over two thirds of bonded stake, then the trie path to the post-state hash.	NEEDS NODE N1
replayed	Fetch the process by hash and its inputs, and run it locally.	NEEDS ISPACE ADAPTER

Against mock nodes: a lying observer is outvoted, a split is an error, a rollback is stale, and nothing is deployed before the user consents.



TEST EVIDENCE

61 WORKSPACE TESTS, AS REPORTED

crate	tests	what they establish
gaze-exec	8	The lamp; identical replay hashes; confinement; runaway pages bounded
gaze-dom-core, graded, knf	16	The protocol engine, the semirings, the container
gaze-dom-blitz	7	Blitz and MemDom commit identical hashes; scoped queries cannot leak
gaze-broker	5	Policy, prompts, remembered grants, routes, revocation
gaze-net, store, blob	10	No ambient credentials; torn-tail recovery; verified cache
gaze-shard	11	Deploys verify; quorum, freshness and consent hold
gaze-reach	3	Wasm commits exactly the native hashes
gaze-shell	1	Settings



UPSTREAM WORK ON CAMPF1R3

BRANCH GAZE/K1G-APERTURES, REV C4E9271, ALSO CARRIED AS A PATCH

Work packages landed

- **U1** ground data at level K1G: booleans, 64-bit integers, strings, bytes, lists, tuples and maps, end to end
- **U3** a machine generic over its observer, so the page meter can be installed
- **U4** apertures: bind, take outbound, inject
- **U6** the keyed minter

Defects found and fixed

- Substitution reversed the arguments of a polyadic send
- The matcher bound a name pattern to the raw quote instead of applying $@^*x = x$, so a received name could not be sent on
- A quoted pattern $@\{^*y\}$ could not match an unforgeable name



RELEASE AND REACH

platform	artifacts	signing
macOS	Universal .dmg, arm64 and x86_64	Developer ID; hardened runtime, no JIT exception; notarised
Windows	x64 .msi and portable .zip	Authenticode, SHA-256, RFC 3161 timestamp
Linux	.deb, .ApplImage, .tar.gz	GPG-signed SHA256SUMS over every platform's artifacts

No release tagged yet

Pushing a vX.Y.Z tag builds a draft release. Signing needs the certificate and key secrets; without them each platform builds unsigned.

Reach tier

Pages run in stock browsers through one wasm module and a small JavaScript host. The shard is a dead channel there: no key custody a page cannot reach.



GATE STATUS

THE PROPOSAL'S FIVE GATES, READ AGAINST THE CODE

gate	criterion	status
G0	A K1 program changes rendered text in Blitz; measurements written down	Program met. Frame-time and wasm-size measurements are not recorded
G1	TodoMVC in f1r3lang; a recorded session replays identically	Replay met. No TodoMVC in the repository
G2	A site on a local three-validator shard survives a hostile proxy	Shown against mock nodes, not yet a real shard
G3	A page using a sampled clause replays from its log	Not started: waits on U2 and U5
G4	Public alpha	Installer pipeline built; no tag; legacy tab, gateway and chrome as a f1r3lang page outstanding



DEVIATIONS FOR REVIEW

FLAGGED BY THE IMPLEMENTATION AGAINST THE SPECIFICATION

Authority of names

A name reached through another inherits its root, so parent works; a new attenuate verb confines components.

Ground values

Two node variants, Lit and Coll, rather than one Ground(Lit).

Commit hash

Hashes the committed document per frame, not the write batch.

One executive per window

Tabs run on the window thread, paced 16 ms foreground, 250 ms background; no worker per tab.

Node HTTP API

The bridge speaks HTTP, not Embers' gRPC client, to avoid protoc at build time. Deploys are byte-identical.

Keys on Linux

A 0600 file in the profile; macOS and Windows use the OS keychain.



NOT YET BUILT

Graded resolver

U2 guards and U5 resolver seam. K2 and graded pages are refused at load; gaze-graded's semirings are ready to wire in.

Wallet

Decided: the agent driving the browser pays for node work, reusing the F1R3Sky wallet. The code has keys and signed deploys, no wallet.

Proof and replayed rungs

Wait on N1 in f1r3node-rust and on the campf1r3-ospace adapter.

rho:serve resolution

Depends on VersionedRegistry step 6 on the node.

Legacy, gateway, devtools

Servo tab, projection gateway, time travel, chrome as a f1r3lang page.

Map rest patterns

Event patterns must list every field. The CampF1R3 remainder-pattern branch would lift this.



OPEN DECISIONS AND OBLIGATIONS

Decisions still open

- Compile surface f1r3lang in the browser, or ship kernel normal forms only (v1 keeps it out)
- HTML, or a document language defined in the f1r3lang DDL
- Servo or CEF for the legacy tab
- End-to-end encryption or session-host confinement for private sessions

Obligations

- Build the campf1r3-ospace adapter and test replay of rspace++ logs
- Round-mode resolution over contention components
- Settle whether K1G encoding agrees with the node's Expr hashes
- Specify passkey delegation: a P-256 assertion authorising a secp256k1 session key
- Measure scene transport for page processes on all three platforms



PROPOSED NEXT STEPS

IN SUGGESTED ORDER

- | | | | |
|---|--|---|--|
| 1 | Review the six deviations
Fold the accepted ones into a specification v0.2 and the DOM protocol document. | 2 | Close G0 and G1
Record the frame-time and wasm-size measurements; write TodoMVC in f1r3lang. |
| 3 | Integrate the F1R3Sky wallet
So the agent driving the browser pays for node work, and Linux keys leave the profile file. | 4 | Run G2 against a real shard
A local three-validator shard behind a hostile proxy; start N1 on f1r3node-rust. |
| 5 | Land U2 and U5
Wire gaze-graded into the executive to reach G3. | 6 | Cut a first signed build
Load the signing secrets and tag v0.1.0 to prove the P4 pipeline end to end. |





F1R3FLY.IO

SOURCES

github.com/F1R3FLY-io/F1R3Gaze

github.com/F1R3FLY-io/publications/f1r3gaze

github.com/F1R3FLY-io/publications/fuzzyware

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